

Python Programming

Class Assignment 2

Python and NumPy

Problems

Problem 1. Make

Make a numpy array and print it

Problem 2. Temperature Analysis

The temperatures recorded over seven days are:

```
[32.5, 31.8, 30.2, 33.1, 34.5, 29.8, 31.0]
```

Store the values in a NumPy array.

Determine:

- Average temperature
- Maximum temperature
- Minimum temperature
- Day with the highest temperature
- All temperatures above the average

Write a function that accepts the temperature array and returns the required analysis.

Problem 3. Working with a Matrix

Create the following NumPy array:

```
1  2  3  4
5  6  7  8
9 10 11 12
```

Without creating another matrix manually, extract:

- The first row
- The last row
- The first column
- The last column
- The middle 2×2 section
- Every alternate element

Also determine its shape, number of dimensions, and total number of elements.

Problem 4. Even and Odd Numbers

Generate an array containing the integers from 1 to 50 selected randomly.

Create two new arrays:

- One containing only even numbers
- One containing only odd numbers

Do this using NumPy operations rather than manually constructing the arrays.

Problem 5. Numbers Above a Threshold

Generate 30 random integers between 1 and 100.

Write a program that:

- Displays the generated array
- Finds all values greater than 50
- Finds all values between 25 and 75
- Counts how many values are greater than 50
- Calculates the average of the values greater than 50

Problem 6. Student Performance Matrix

Consider five students and three subjects:

	Mathematics	Physics	Programming
Student 1	78	82	91
Student 2	65	74	80
Student 3	92	88	95
Student 4	56	69	72
Student 5	84	90	87

Store only the numerical marks in a NumPy array.

Determine:

- Average marks of each student
- Average marks in each subject
- Highest mark in the entire dataset
- Lowest mark in the entire dataset
- Student with the highest average
- Subject with the highest average

Problem 7. Matrix Operations

Create two 3×3 NumPy matrices.

Perform:

- Addition
- Subtraction
- Element-wise multiplication
- Matrix multiplication

Compare the results of element-wise multiplication and matrix multiplication.

In your explanation, describe why the two operations produce different results.

Problem 8. Scientific Data Transformation

Suppose an experiment produces the following measurements:

[2, 4, 6, 8, 10, 12]

Create a NumPy array and calculate:

- The square of every measurement
- The square root of every measurement
- The exponential of every measurement

Then calculate the mean of the transformed data.

Write your solution as a function that accepts an array as input.

Problem 9. Reshaping Experimental Data

Generate the numbers from 1 to 24 using NumPy.

Reshape the resulting array into:

- A 4×6 array
- A 6×4 array
- A $2 \times 3 \times 4$ array

Display the shape of each resulting array.

Then convert the 4×6 array back into a one-dimensional array.

Problem 10. Data Combination

Create two NumPy arrays representing measurements from two experiments.

For example:

```
experiment_1 = [10, 20, 30, 40, 50]
experiment_2 = [15, 25, 35, 45, 55]
```

Combine the data in different ways so that you can obtain:

- A single one-dimensional array containing all measurements
- A matrix where the two experiments form separate rows
- A matrix where the two experiments form separate columns

Determine the shape of each result.

Problem 11. Simple Data Cleaning

Consider the following experimental measurements:

```
[10, 12, 15, 200, 13, 11, 14, 250, 12, 15]
```

Assume that values greater than 100 are invalid measurements.

Write a program that:

- Identifies the invalid measurements
- Extracts only the valid measurements
- Calculates the mean of the valid measurements
- Calculates the minimum and maximum valid measurements

Do not modify the original array.

Problem 12. Image-like Matrix

Consider the following matrix:

```
image = np.array([
    [0, 0, 0, 255, 255],
    [0, 0, 128, 255, 255],
    [0, 128, 128, 128, 255],
    [0, 0, 128, 0, 0],
    [255, 0, 0, 0, 255]
])
```

Treat the values as pixel intensities.

Determine:

- Minimum pixel intensity
- Maximum pixel intensity
- Mean pixel intensity
- Number of pixels with intensity greater than 100
- All pixels with intensity equal to 255

Then create a new array containing only pixels with intensity greater than 100.

Problem 13. Experiment Comparison

Two experiments produced the following measurements:

```
experiment_A = [12, 15, 14, 18, 20, 17, 16]
experiment_B = [10, 13, 15, 16, 19, 21, 18]
```

Using NumPy, determine:

- Mean of each experiment
- Standard deviation of each experiment
- Maximum value of each experiment
- Minimum value of each experiment
- Difference between the two experiments for each measurement

Determine which experiment shows greater variation in its measurements.

Problem 14. Mini Data Analysis Challenge

Generate a 10×5 NumPy array representing measurements collected from 10 experiments across 5 variables.

The values should be randomly generated.

Your program should:

- Display the dataset
- Display its shape and dimensions
- Calculate the mean of each variable
- Calculate the mean of each experiment
- Find the maximum and minimum value in the entire dataset
- Identify all values above the overall mean
- Reshape the data into a different valid shape
- Flatten the reshaped data back into one dimension

Organize your solution using at least two functions.

Problem 15. Final Challenge: Student Performance Analysis

Create a complete student performance analysis program.

Consider:

- 10 students
- 4 subjects
- Marks between 0 and 100

Generate the marks using NumPy.

Your program should:

- (a) Store the marks in a two-dimensional NumPy array.
- (b) Calculate the average mark of every student.
- (c) Calculate the average mark of every subject.
- (d) Identify the student with the highest average.
- (e) Identify the student with the lowest average.
- (f) Identify the highest mark obtained in the entire dataset.
- (g) Identify the lowest mark obtained in the entire dataset.
- (h) Count how many students have an average above 75.
- (i) Extract all marks greater than 90.
- (j) Assign a grade to each student's average using conditional statements.
- (k) Store the final student information in an appropriate Python data structure.
- (l) Write functions wherever appropriate.

The final output should present the analysis in a clear and readable format.

Submission

Submit:

- The completed Jupyter Notebook.